

Nomalungelo Mdleko

Practical 2: SQL JOINS

Question 1

PRACTICAL_2.SQL_JOINSettingsOpen in WorkspacesCode Versions

```
1 --INNER JOIN
2 --List all orders along with the customer name and product name.
3 SELECT O.ORDERID, O.ORDERDATE, C.CUSTOMERNAME, P.PRODUCTNAME,O.QUANTITY
4 FROM ORDERS_LARGE AS O
5 INNER JOIN CUSTOMERS_LARGE AS C
6 ON O.CUSTOMERID = C.CUSTOMERID
7 INNER JOIN PRODUCT_LARGE AS P
```

ResultsChart

	# ORDERID	ORDERDATE	CUSTOMERNAME	PRODUCTNAME	QUANTITY
1	1	2023-06-10	Customer_1251	Product_2014	10
2	2	2023-12-07	Customer_1236	Product_2004	5
3	3	2024-10-26	Customer_1170	Product_2171	9
4	4	2023-02-17	Customer_1344	Product_2007	2
5	5	2024-11-06	Customer_1319	Product_2061	2
6	6	2024-11-23	Customer_1185	Product_2190	3
7	7	2023-07-29	Customer_1011	Product_2099	8
8	8	2023-12-06	Customer_1322	Product_2078	7
9	9	2025-01-25	Customer_1224	Product_2043	7
10	10	2023-07-19	Customer_1010	Product_2141	3
11	11	2024-11-21	Customer_1324	Product_2090	1
12	12	2023-06-19	Customer_1248	Product_2115	6

Question 2

9

```
10 --Which customers have placed at least one order?
11 SELECT C.CUSTOMERID, C.CUSTOMERNAME, C.COUNTRY, O.ORDERID, O.ORDERDATE
12 FROM CUSTOMERS_LARGE AS C
13 INNER JOIN ORDERS_LARGE AS O
14 ON C.CUSTOMERID = O.CUSTOMERID;
15
```

ResultsChart

	# CUSTOMERID	CUSTOMERNAME	COUNTRY	# ORDERID	ORDERDATE
1	1251	Customer_1251	Germany	1	2023-06-10
2	1236	Customer_1236	Australia	2	2023-12-07
3	1170	Customer_1170	Germany	3	2024-10-26
4	1344	Customer_1344	Canada	4	2023-02-17
5	1319	Customer_1319	USA	5	2024-11-06
6	1185	Customer_1185	Australia	6	2024-11-23
7	1011	Customer_1011	Germany	7	2023-07-29
8	1322	Customer_1322	Australia	8	2023-12-06
9	1224	Customer_1224	Australia	9	2025-01-25
10	1010	Customer_1010	UK	10	2023-07-19
11	1324	Customer_1324	India	11	2024-11-21
12	1248	Customer_1248	Germany	12	2023-06-19

Question 3

PRACTICAL_2.SQL_JOIN Settings Open in Workspaces Code Versions

```

18 SELECT C.CustomerID, C.CustomerName, C.Country, O.OrderID, O.OrderDate, O.ProductID, O.Quantity
19 FROM CUSTOMERS_LARGE AS C
20 LEFT JOIN ORDERS_LARGE AS O
21 ON C.CUSTOMERID = O.CUSTOMERID;
22
23 --List all products and how many times each was ordered (if any).

```

Results Chart

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	ORDERDATE	# PRODUCTID	# QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2
6	1185	Customer_1185	Australia	6	2024-11-23	2190	3
7	1011	Customer_1011	Germany	7	2023-07-29	2099	8
8	1322	Customer_1322	Australia	8	2023-12-06	2078	7
9	1224	Customer_1224	Australia	9	2025-01-25	2043	7
10	1010	Customer_1010	UK	10	2023-07-19	2141	3
11	1324	Customer_1324	India	11	2024-11-21	2090	1
12	1248	Customer_1248	Germany	12	2023-06-19	2115	6

Question 4

PRACTICAL_2.SQL_JOIN Settings Open in Workspaces Code Versions

```

23 --List all products and how many times each was ordered (if any).
24 SELECT P.PRODUCTID, P.PRODUCTNAME,
25 COUNT(O.ORDERID) AS TOTAL_ORDERS
26 FROM PRODUCT_LARGE AS P
27 LEFT JOIN ORDERS_LARGE AS O
28 ON P.PRODUCTID = O.PRODUCTID
29 GROUP BY P.PRODUCTID, P.PRODUCTNAME;

```

Results Chart

	# PRODUCTID	A PRODUCTNAME	# TOTAL_ORDERS
1	2171	Product_2171	15
2	2177	Product_2177	20
3	2073	Product_2073	19
4	2089	Product_2089	20
5	2054	Product_2054	24
6	2019	Product_2019	17
7	2190	Product_2190	20
8	2119	Product_2119	22
9	2182	Product_2182	17
10	2042	Product_2042	22
11	2169	Product_2169	18
12	2038	Product_2038	16

Question 5

```

33 SELECT O.ORDERID, O.ORDERDATE, P.PRODUCTID, P.PRODUCTNAME, P.PRICE, O.QUANTITY
34 FROM ORDERS_LARGE AS O
35 RIGHT JOIN PRODUCT_LARGE AS P
36 ON O.PRODUCTID = P.PRODUCTID;
37

```

#	ORDERID	ORDERDATE	PRODUCTID	PRODUCTNAME	PRICE	QUANTITY
1	1	2023-06-10	2014	Product_2014	522	10
2	2	2023-12-07	2004	Product_2004	1996	5
3	3	2024-10-26	2171	Product_2171	76	9
4	4	2023-02-17	2007	Product_2007	156	2
5	5	2024-11-06	2061	Product_2061	1595	2
6	6	2024-11-23	2190	Product_2190	1755	3
7	7	2023-07-29	2099	Product_2099	1674	8
8	8	2023-12-06	2078	Product_2078	333	7
9	9	2025-01-25	2043	Product_2043	1947	7
10	10	2023-07-19	2141	Product_2141	1599	3
11	11	2024-11-21	2090	Product_2090	1662	1
12	12	2023-06-19	2115	Product_2115	1585	6

Question 6

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37
38 --Which customers have made orders, and include customers even if they have never placed an order.
39 SELECT C.CUSTOMERID, C.CUSTOMERNAME, C.COUNTRY, O.ORDERID, O.ORDERDATE, O.PRODUCTID, O.QUANTITY
40 FROM ORDERS_LARGE AS O
41 RIGHT JOIN CUSTOMERS_LARGE AS C
42 ON O.CUSTOMERID = C.CUSTOMERID;
43

```

#	CUSTOMERID	CUSTOMERNAME	COUNTRY	ORDERID	ORDERDATE	PRODUCTID	QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2
6	1185	Customer_1185	Australia	6	2024-11-23	2190	3
7	1011	Customer_1011	Germany	7	2023-07-29	2099	8
8	1322	Customer_1322	Australia	8	2023-12-06	2078	7
9	1224	Customer_1224	Australia	9	2025-01-25	2043	7
10	1010	Customer_1010	UK	10	2023-07-19	2141	3
11	1324	Customer_1324	India	11	2024-11-21	2090	1
12	1248	Customer_1248	Germany	12	2023-06-19	2115	6

Question 7

PRACTICAL_2.SQL_JOIN Settings Open in Workspaces Code Versions

```

45 --List all customers and orders, showing NULLs where customers have not ordered or where orders have
46 no customer info.
47 SELECT C.CUSTOMERID, C.CUSTOMERNAME, C.COUNTRY, O.ORDERID, O.ORDERDATE, O.PRODUCTID, O.QUANTITY
48 FROM CUSTOMERS_LARGE AS C
49 FULL OUTER JOIN ORDERS_LARGE AS O
50 ON C.CUSTOMERID = O.CUSTOMERID;

```

Results Chart

	# CUSTOMERID	A CUSTOMERNAME	A COUNTRY	# ORDERID	ORDERDATE	# PRODUCTID	# QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2
6	1185	Customer_1185	Australia	6	2024-11-23	2190	3
7	1011	Customer_1011	Germany	7	2023-07-29	2099	8
8	1322	Customer_1322	Australia	8	2023-12-06	2078	7
9	1224	Customer_1224	Australia	9	2025-01-25	2043	7
10	1010	Customer_1010	UK	10	2023-07-19	2141	3
11	1324	Customer_1324	India	11	2024-11-21	2090	1
12	1248	Customer_1248	Germany	12	2023-06-19	2115	6

Question 8

PRACTICAL_2.SQL_JOIN Settings Open in Workspaces Code Versions

```

51 --List all products and orders, showing NULLs where products were never ordered or orders are missing
52 product info.
53 SELECT P.PRODUCTID, P.PRODUCTNAME, P.PRICE, O.ORDERID, O.ORDERDATE, O.CUSTOMERID, O.QUANTITY
54 FROM PRODUCT_LARGE AS P
55 FULL OUTER JOIN ORDERS_LARGE AS O
ON P.PRODUCTID = O.PRODUCTID;

```

Results Chart

	# PRODUCTID	A PRODUCTNAME	# PRICE	# ORDERID	ORDERDATE	# CUSTOMERID	# QUANTITY
1	2014	Product_2014	522	1	2023-06-10	1251	10
2	2004	Product_2004	1996	2	2023-12-07	1236	5
3	2171	Product_2171	76	3	2024-10-26	1170	9
4	2007	Product_2007	156	4	2023-02-17	1344	2
5	2061	Product_2061	1595	5	2024-11-06	1319	2
6	2190	Product_2190	1755	6	2024-11-23	1185	3
7	2099	Product_2099	1674	7	2023-07-29	1011	8
8	2078	Product_2078	333	8	2023-12-06	1322	7
9	2043	Product_2043	1947	9	2025-01-25	1224	7
10	2141	Product_2141	1599	10	2023-07-19	1010	3
11	2090	Product_2090	1662	11	2024-11-21	1324	1
12	2115	Product_2115	1585	12	2023-06-19	1248	6